

DS-GA 1018: Homework 5

Due Monday December 2nd at 5:00 pm

Problem 1 (10 points): Imagine that we are doing GP regression using the kernel function:

$$\kappa(x, x') = \cos\left(\frac{2\pi|x - x'|}{P}\right). \quad (1)$$

We are also making the default choice for the mean function: $\mu(x) = 0$. We have two observations at x_1 and x_2 with values y_1 and y_2 . Also assume that:

$$x_2 = x_1 + P. \quad (2)$$

Finally, assume that our model includes independent, Gaussian observation noise with variance σ_w^2 . Using this GP and observations, answer the following questions:

i. (5 points): Derive the mean and variance for a new observation x_* . Write your answer in terms of:

$$k_* = \kappa(x_*, x_1). \quad (3)$$

For a Gaussian Process with observations $\mathbf{y} = [y_1, y_2]^\top$ at x_1 and x_2 , we know from the lecture notes that :

$$\begin{bmatrix} \mathbf{y} \\ y_* \end{bmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix} \mathbf{K} + \sigma_w^2 \mathbf{I} & \mathbf{K}_* \\ \mathbf{K}_*^\top & \mathbf{K}_{**} \end{bmatrix}\right)$$

where: $\mathbf{K} = \begin{bmatrix} \kappa(x_1, x_1) & \kappa(x_1, x_2) \\ \kappa(x_2, x_1) & \kappa(x_2, x_2) \end{bmatrix}$. $\mathbf{K}_* = [\kappa(x_*, x_1), \kappa(x_*, x_2)]^\top$ and $\mathbf{K}_{**} = \kappa(x_*, x_*)$. The mean $\mu(x_*)$ and variance $\sigma^2(x_*)$ of the prediction at x_* are given by:

$$\begin{aligned} \mu(x_*) &= \mathbf{K}_*^\top (\mathbf{K} + \sigma_w^2 \mathbf{I})^{-1} \mathbf{y} \\ \sigma^2(x_*) &= \mathbf{K}_{**} - \mathbf{K}_*^\top (\mathbf{K} + \sigma_w^2 \mathbf{I})^{-1} \mathbf{K}_* \end{aligned}$$

Now, given that $x_1 = x_2 + P$, we know $|x_1 - x_2| = P$, thus we have:

$\mathbf{K} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $\mathbf{K}_* = \begin{bmatrix} k_* \\ \cos\left(\frac{2\pi(x_* - x_2)}{P}\right) \end{bmatrix}$ and $\mathbf{K}_{**} = \kappa(x_*, x_*) = 1$. Also we know that $\cos\left(\frac{2\pi|x_* - x_2|}{P}\right) = \cos\left(\frac{2\pi|x_* - x_1 - P|}{P}\right) = \cos\left(\frac{2\pi|x_* - x_1|}{P}\right) = k_*$ fitting them into the equations above and we get the final answer:

$$\begin{aligned} \mu_* &= \mathbf{K}_*^\top (\mathbf{K} + \sigma_w^2 \mathbf{I})^{-1} \mathbf{y} \\ &= \frac{1}{\sigma_w^2 (2 + \sigma_w^2)} \cdot \begin{bmatrix} k_* & k_* \end{bmatrix} \begin{bmatrix} 1 + \sigma_w^2 & -1 \\ -1 & 1 + \sigma_w^2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \\ &= \frac{k_*}{\sigma_w^2 (2 + \sigma_w^2)} [(1 + \sigma_w^2)(y_1 + y_2) - (y_1 + y_2)] \\ &= \frac{k_*}{\sigma_w^2 (2 + \sigma_w^2)} [\sigma_w^2 (y_1 + y_2)] \\ &= \frac{k_* (y_1 + y_2)}{2 + \sigma_w^2} \end{aligned}$$

and that:

$$\begin{aligned} \Sigma_* &= 1 - \frac{2\sigma_w^2 k_*^2}{\sigma_w^2 (2 + \sigma_w^2)} \\ &= \frac{2 + \sigma_w^2 - 2k_*^2}{2 + \sigma_w^2} \end{aligned}$$

ii. (2 points): Now assume that $x_\star = x_1 + 3P$. Substitute in the correct value of $k_\star$ and write the mean and variance prediction for the observation at $x_\star$.

All the derivation still holds from (i), given $x_\star = x_1 + 3P$, we have that the kernel function:

$$\begin{aligned}\kappa(x_\star, x_1) &= \cos\left(\frac{2\pi(x_1 + 3P - x_1)}{P}\right) = \cos(6\pi) = 1 \\ \kappa(x_\star, x_2) &= \cos\left(\frac{2\pi(x_1 + 3P - (x_1 + P))}{P}\right) = \cos(4\pi) = 1\end{aligned}$$

Thus, $\mathbf{K}_\star = [1, 1]^\top$, all the other remain the same as (i), and we could get:

$$\begin{aligned}\mu(x_\star) &= \frac{y_1 + y_2}{2 + \sigma_w^2} \\ \Sigma(x_\star) &= 1 - \frac{2}{2 + \sigma_w^2} = \frac{\sigma_w^2}{2 + \sigma_w^2}\end{aligned}$$

iii. (3 points): Still assuming that $x_\star = x_1 + 3P$, take the limit of $\sigma_w \rightarrow 0$. What is the new answer for the mean and variance? Explain the intuition behind this result.

When $\sigma_w \rightarrow 0$, we have:

$$\begin{aligned}\mu(x_\star) &\rightarrow \frac{y_1 + y_2}{2} \\ \Sigma(x_\star) &\rightarrow 0\end{aligned}$$

The intuition would be that since the noise is assumed to be zero, the prediction at the test point $x_\star$ is totally dependent on the observations themselves. It's like there is no influence from the prior.

Problem 2 (10 points): In this problem we will prove a property we took advantage of in class:

$$\int_0^P \left[\sum_{n=0}^{\infty} A_n \cos \left(\frac{2\pi n t}{P} \right) \right] \cos \left(\frac{2\pi m t}{P} \right) dt = \frac{A_m P}{2} \quad (4)$$

$$\int_0^P \left[\sum_{n=0}^{\infty} B_n \sin \left(\frac{2\pi n t}{P} \right) \right] \cos \left(\frac{2\pi m t}{P} \right) dt = 0, \quad (5)$$

where m is a positive integer. In this problem, we are going to prove these properties.

i. (4 points):

Show that the following integral:

$$\int_0^P \cos\left(\frac{2\pi nt}{P}\right) \cos\left(\frac{2\pi mt}{P}\right) dt \quad (6)$$

gives $\frac{P}{2}$ if $m = n$ and 0 otherwise. *Hint: Take advantage of the identity $\cos(\alpha)\cos(\beta) = \frac{1}{2}(\cos(\alpha + \beta) + \cos(\alpha - \beta))$. Depending on how you solve the problem, you may also need to use l'hospital's rule.*

We will have the following derivations:

1. If $m = n$, we have

$$\begin{aligned} & \int_0^P \cos\left(\frac{2\pi mt}{P}\right) \cos\left(\frac{2\pi mt}{P}\right) dt \\ &= \int_0^P \frac{1}{2} \left(\cos \frac{4\pi mt}{P} + \cos 0 \right) dt \\ &= \int_0^P \left(\frac{1}{2} + \frac{1}{2} \cos \frac{4\pi mt}{P} \right) dt \\ &= \frac{1}{2}P + \left(\frac{1}{2} \sin \frac{4\pi mt}{P} \cdot \frac{P}{4\pi m} \right) \Big|_0^P \\ &= \frac{1}{2}P + \frac{1}{2}(\sin 4\pi m - \sin 0) \\ &= \frac{1}{2}P \end{aligned}$$

2. If $m \neq n$, we will have

$$\begin{aligned} & \int_0^P \cos\left(\frac{2\pi nt}{p}\right) \cos\left(\frac{2\pi mt}{p}\right) dt \\ &= \int_0^P \frac{1}{2} \left\{ \cos\left(\frac{2\pi(m+n)t}{p}\right) + \cos\left(\frac{2\pi(n-m)t}{p}\right) \right\} \\ &= \frac{1}{2} \left\{ \sin\left(\frac{2\pi(m+n)t}{p}\right) \cdot \frac{p}{2\pi(m+n)} + \sin\left(\frac{2\pi(n-m)t}{p}\right) \cdot \frac{p}{2\pi(n-m)} \right\} \Big|_0^P \\ &= \frac{1}{2} \{0 + 0\} \\ &= 0 \end{aligned}$$

ii. (4 points): Show that the following integral:

$$\int_0^P \sin\left(\frac{2\pi nt}{P}\right) \cos\left(\frac{2\pi mt}{P}\right) dt \quad (7)$$

gives 0 for all values of m . *Hint: Take advantage of the identity $\cos(\alpha) \sin(\beta) = \frac{1}{2}(\sin(\alpha + \beta) - \sin(\alpha - \beta))$.*

We will have the following derivations:

1. If $m \neq n$, we have

$$\begin{aligned} & \int_0^P \sin\left(\frac{2\pi nt}{P}\right) \cos\left(\frac{2\pi mt}{P}\right) dt \\ &= \int_0^P \frac{1}{2} \left\{ \sin\left(\frac{2\pi(m+n)t}{P}\right) - \sin\left(\frac{2\pi(m-n)t}{P}\right) \right\} dt \\ &= \frac{1}{2} \left\{ -\cos\left(\frac{2\pi(m+n)t}{P}\right) \cdot \frac{P}{2\pi(m+n)} + \cos\left(\frac{2\pi(m-n)t}{P}\right) \cdot \frac{P}{2\pi(m-n)} \right\} \Big|_0^P \\ &= \frac{1}{2} \left\{ -\frac{P}{2\pi(m+n)} + \frac{P}{2\pi(m-n)} - \left(-\frac{P}{2\pi(m+n)} + \frac{P}{2\pi(m-n)} \right) \right\} \\ &= 0 \end{aligned}$$

2. If $m = n$, we have

$$\begin{aligned} & \int_0^P \sin\left(\frac{2\pi mt}{P}\right) \cos\left(\frac{2\pi mt}{P}\right) dt \\ &= \int_0^P \frac{1}{2} \left\{ \sin\left(\frac{2\pi(m+m)t}{P}\right) - \sin\left(\frac{2\pi(m-m)t}{P}\right) \right\} dt \\ &= \frac{1}{2} \left\{ -\cos\left(\frac{2\pi(m+n)t}{P}\right) \cdot \frac{P}{2\pi(m+n)} \right\} \Big|_0^P \\ &= \frac{1}{2} \left\{ -\frac{P}{2\pi(m+n)} - \left(-\frac{P}{2\pi(m+n)} \right) \right\} \\ &= 0 \end{aligned}$$

iii. (2 points): Combine the results from the previous two subproblems to derive Equation 4 and Equation 5.

1. For equation 4, we will have

$$\begin{aligned}
 & \int_0^P \left[\sum_{n=0}^{\infty} A_n \cos \left(\frac{2\pi nt}{P} \right) \right] \cos \left(\frac{2\pi mt}{P} \right) dt \\
 &= A_m \int_0^P \cos \left(\frac{2\pi mt}{P} \right) \cos \left(\frac{2\pi mt}{P} \right) dt + \int_0^P \left[\sum_{n \neq m}^{\infty} A_n \cos \left(\frac{2\pi nt}{P} \right) \right] \cos \left(\frac{2\pi mt}{P} \right) dt \\
 &= \frac{A_m P}{2}
 \end{aligned}$$

2. For equation 5, by (ii) we could directly know that

$$\begin{aligned}
 & \int_0^P \left[\sum_{n=0}^{\infty} B_n \sin \left(\frac{2\pi nt}{P} \right) \right] \cos \left(\frac{2\pi mt}{P} \right) dt \\
 &= 0
 \end{aligned}$$

Problem 3 (6 points): One of the counter-intuitive properties of the Fourier transform is that a signal that is sparse in the time-domain will be dense in the frequency domain. We will show that explicitly in this problem. For each sub-problem, assume we have a sequence of observations $\{X_0, \dots, X_{T-1}\}$ with the observations evenly spaced out in time.

i. (2 points): Assume $X_n = 1$ for some n and $X_t = 0$ for $t \neq n$. What is the discrete Fourier transform of the signal?

The signal is a single pulse at index n . Thus the DFT for this signal is given by:

$$X_k = \frac{1}{T} \sum_{t=0}^{T-1} X_t e^{-2\pi i k t / T}$$

Since $X_t = 0$ for $t \neq n$, the sum reduces to:

$$X_k = \frac{1}{T} X_n e^{-2\pi i k n / T} = \frac{1}{T} e^{-2\pi i k n / T}$$

ii. (4 points): Assume $X_n = 1, X_{T-n} = 1$ for some $n > 0$ and $X_t = 0$ for $t \neq n, T - n$. What is the discrete Fourier transform of the signal? Simplify your answer. *Hint: $\sin(x) = -\sin(-x) = -\sin(2\pi j - x)$ for any integer j . Similarly, $\cos(x) = \cos(-x) = \cos(2\pi j - x)$ for any integer j .*

The signal is symmetric about $T/2$, with non-zero values at n and $T - n$. The DFT is computed as:

$$X_k = \frac{1}{T} \sum_{t=0}^{T-1} X_t e^{-2\pi i k t / T}$$

and this sum reduces to:

$$X_k = \frac{1}{T} \left(e^{-2\pi i k n / T} + e^{-2\pi i k (T-n) / T} \right)$$

We then simplify the second term:

$$e^{-2\pi i k (T-n) / T} = e^{-2\pi i k} e^{2\pi i k n / T} = e^{2\pi i k n / T} \quad \left(\text{since } e^{-2\pi i k} = 1 \text{ for integers } k \right)$$

Then we could combine terms:

$$X_k = \frac{1}{T} \left(e^{-2\pi i k n / T} + e^{2\pi i k n / T} \right)$$

Finally, this simplifies to:

$$X_k = \frac{2}{T} \cos \left(\frac{2\pi k n}{T} \right)$$

Thus, the DFT is a real-valued cosine function with amplitude 2 and frequency dependent on n and T .